

Nexum Suum – Make-Up Water System Troubleshooting SOP

■ Page1:FlowFailure, ValveMalfunction,HardnessControl,OverfillProtection

If System Receives No Make-Up Water:

- Check Float Valve, Solenoid, or BAS Actuator Position.
- Verify Isolation Valve is Open and Strainer is Clean.
- Confirm City Supply Pressure is Within Operating Range.

If Make-Up Valve Fails Open and System Overfills:

- Inspect Float or Level Sensor for Mechanical Sticking.
- Test Solenoid for Power Drop or Seal Failure.
- Add Backup High-Level Alarm for Overflow Protection.

If Water Hardness or Conductivity is High:

- Confirm Softener or RO System is Active and Regenerating.
- Check Brine Tank Level, Timer, and Salt Bridge. Validate
- Bypass Valve is Not Accidentally Open.

If Water Temperature or Pressure Fluctuates:

- Review Mixing Valve Setting or Seasonal Supply Temp Change.
- Inspect PRV and Check Valve Orientation and Operation.

Nexum Suum – Make-Up Water System Troubleshooting SOP (Cont'd)

■ Page2:Symptom-to-Action Mapping+Real-WorldThresholds

Symptom	Action
No Water Reaching System	Check float/solenoid valve, verify pressure and isolation.
System Overfilling	Inspect float sensor, test solenoid seal or actuator fault.
Hardness Too High	Review softener regen cycle, salt level, bypass settings.
Fluctuating Pressure/Temp	Inspect PRV, mixing valve, seasonal effects on source.

■ Operating Thresholds:

- Water Pressure Supply: 40–80 psi (municipal)
- Target Fill Pressure: 12–25 psi (based on system elevation)
- Hardness Post-Softener: <1 ppm
- Conductivity: <50 $\mu\text{S}/\text{cm}$ for boiler, <1000 $\mu\text{S}/\text{cm}$ for tower
- Brine Tank Salt Level: $\geq 30\%$ full and no salt bridging
- Float Valve Setpoint: 3–6" below overflow
- Solenoid Voltage: 24VDC or 120VAC per spec
- System Make-up Frequency: <1x per hour on stable closed loop